

Exercise 1: Configuring a Basic Spring Application

Source Code

BookRepository.java

```
package com.library.repository;

public class BookRepository {
    public void saveBook() {
        System.out.println("Book Repository is available.");
    }
}
```

BookService.java

```
package com.library.service;

public class BookService {
    public void displayBook() {
        System.out.println("Book Service is working successfully.");
    }
}
```

LibraryManagementApplication.java

```
package com.library;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import com.library.service.BookService;
import com.library.repository.BookRepository;
```

@SpringBootApplication

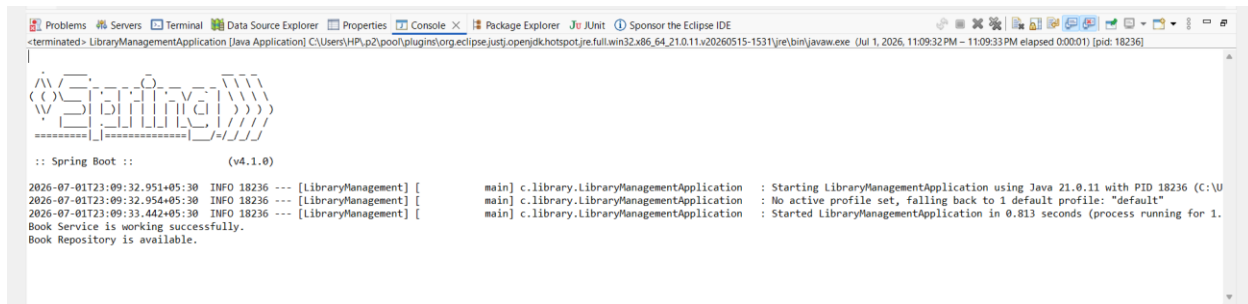
```
public class LibraryManagementApplication {
    public static void main(String[] args) {
        SpringApplication.run(LibraryManagementApplication.class, args);
        BookService service = new BookService();
        BookRepository repository = new BookRepository();

        service.displayBook();
        repository.saveBook();
    }
}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
         http://www.springframework.org/schema/beans
         https://www.springframework.org/schema/beans/spring-beans.xsd">
  <bean id="bookService"
        class="com.library.service.BookService"/>
  <bean id="bookRepository"
        class="com.library.repository.BookRepository"/>
</beans>
```

Output



```
<terminated> LibraryManagementApplication [Java Application] C:\Users\HP\AppData\Local\Temp\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.11\jre\bin\javaw.exe (Jul 1, 2026, 11:09:32 PM - 11:09:33 PM elapsed 0:00:01) [pid: 18236]

:: Spring Boot ::                (v4.1.0)

2026-07-01T23:09:32.951+05:30 INFO 18236 --- [LibraryManagement] [
2026-07-01T23:09:32.954+05:30 INFO 18236 --- [LibraryManagement] [
2026-07-01T23:09:33.442+05:30 INFO 18236 --- [LibraryManagement] [
Book Service is working successfully.
Book Repository is available.

main] c.library.LibraryManagementApplication : Starting LibraryManagementApplication using Java 21.0.11 with PID 18236 (C:\U
main] c.library.LibraryManagementApplication : No active profile set, falling back to 1 default profile: "default"
main] c.library.LibraryManagementApplication : Started LibraryManagementApplication in 0.813 seconds (process running for 1.
```

Result

The LibraryManagement Spring application was successfully configured using Maven and Spring Boot. The BookService and BookRepository classes were created, and the applicationContext.xml file was configured with the required bean definitions. The application executed successfully, displaying "**Book Service is working successfully.**" and "**Book Repository is available.**", confirming that the basic Spring application was configured and executed successfully.

Exercise 2: Implementing Dependency Injection

Source Code

BookRepository.java

```
package com.library.repository;

public class BookRepository {

    public void saveBook() {
        System.out.println("Book Repository is available.");
    }
}
```

```
}  
}
```

BookService.java

```
package com.library.service;  
import com.library.repository.BookRepository;  
public class BookService {  
    private BookRepository bookRepository;  
    public void setBookRepository(BookRepository bookRepository) {  
        this.bookRepository = bookRepository;  
    }  
    public void displayBook() {  
        System.out.println("Book Service is working successfully.");  
        bookRepository.saveBook();  
    }  
}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>  
<beans xmlns="http://www.springframework.org/schema/beans"  
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
    xsi:schemaLocation="  
        http://www.springframework.org/schema/beans  
        https://www.springframework.org/schema/beans/spring-beans.xsd">  
    <bean id="bookRepository"  
        class="com.library.repository.BookRepository"/>  
    <bean id="bookService"  
        class="com.library.service.BookService">  
        <property name="bookRepository" ref="bookRepository"/>  
    </bean>  
</beans>
```

LibraryManagementApplication.java

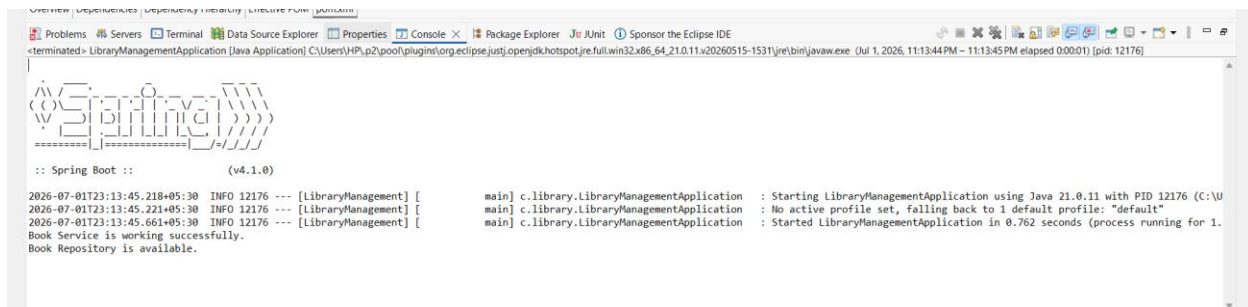
```
package com.library;  
import org.springframework.boot.SpringApplication;  
import org.springframework.boot.autoconfigure.SpringBootApplication;  
import org.springframework.context.ApplicationContext;  
import org.springframework.context.support.ClassPathXmlApplicationContext;
```

```

import com.library.service.BookService;
@SpringBootApplication
public class LibraryManagementApplication {
    public static void main(String[] args) {
        SpringApplication.run(LibraryManagementApplication.class, args);
        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");
        BookService service = (BookService) context.getBean("bookService");
        service.displayBook();
    }
}

```

Output



```

:: Spring Boot ::
      (v4.1.0)

2026-07-01T23:13:45.218+05:30 INFO 12176 --- [LibraryManagement] [
2026-07-01T23:13:45.221+05:30 INFO 12176 --- [LibraryManagement] [
2026-07-01T23:13:45.661+05:30 INFO 12176 --- [LibraryManagement] [
Book Service is working successfully.
Book Repository is available.

main c.library.LibraryManagementApplication : Starting LibraryManagementApplication using Java 21.0.11 with PID 12176 (C:\U
main c.library.LibraryManagementApplication : No active profile set, falling back to 1 default profile: "default"
main c.library.LibraryManagementApplication : Started LibraryManagementApplication in 0.762 seconds (process running for 1.

```

Result

Dependency Injection was implemented successfully using the Spring IoC container and XML configuration. The BookRepository bean was injected into the BookService class through setter injection. The application executed successfully and displayed messages from both the service and repository classes, confirming that dependency injection was configured and working correctly.

Exercise 4: Creating and Configuring a Maven Project

Source Code

pom.xml

```

<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"

    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0

```

```
    https://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.library</groupId>
<artifactId>LibraryManagement</artifactId>
<version>0.0.1-SNAPSHOT</version>
<name>LibraryManagement</name>
<properties>
    <java.version>1.8</java.version>
</properties>
<dependencies>
<dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
    <version>5.3.39</version>
</dependency>
<dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-aop</artifactId>
    <version>5.3.39</version>
</dependency>
<dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-webmvc</artifactId>
    <version>5.3.39</version>
</dependency>
</dependencies>
```

```
<build>

<plugins>

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-compiler-plugin</artifactId>

<version>3.11.0</version>

<configuration>

<source>1.8</source>

<target>1.8</target>

</configuration>

</plugin>

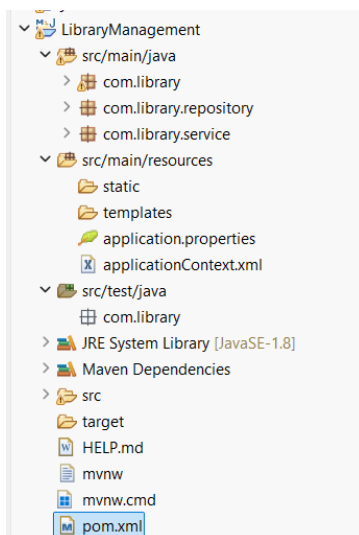
</plugins>

</build>

</project>
```

Output

The Maven project was created and updated successfully, and all required Spring dependencies were downloaded successfully.



Result

The Maven project **LibraryManagement** was created and configured successfully. The required Spring Framework dependencies (Spring Context, Spring AOP, and Spring WebMVC) were added to the pom.xml file, and the Maven Compiler Plugin was configured successfully. The project was updated successfully and is ready for Spring application development.